

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	RAVURU PRANATHI
Register Number	192525076

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *R. Pranathi*

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

- 1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.**

ANS:

Nowadays many companies conduct seasonal sales during festivals like Diwali, Christmas, New Year and other special occasions. During these sales, a large number of customers visit the website at the same time. If the company uses only one or two servers, the website may become slow or even crash because it cannot handle so many requests. Cloud computing provides a better solution through Auto Scaling, Load Balancing and Monitoring Tools.

Auto Scaling is a cloud feature that automatically increases or decreases the number of servers based on the amount of traffic. When many customers visit the website, the cloud automatically adds more servers so that all users can access the website without any delay. After the sale is over and the traffic becomes normal, the extra servers are removed automatically. This helps the company save money because it only pays for the resources that are actually used.

Load Balancing is another important feature of cloud computing. It distributes user requests equally among all available servers. Instead of sending all requests to one server, it shares the workload with multiple servers. This prevents any one server from becoming overloaded. Even if one server fails, the load balancer immediately sends the requests to another working server. Because of this, the website continues to work without interruption.

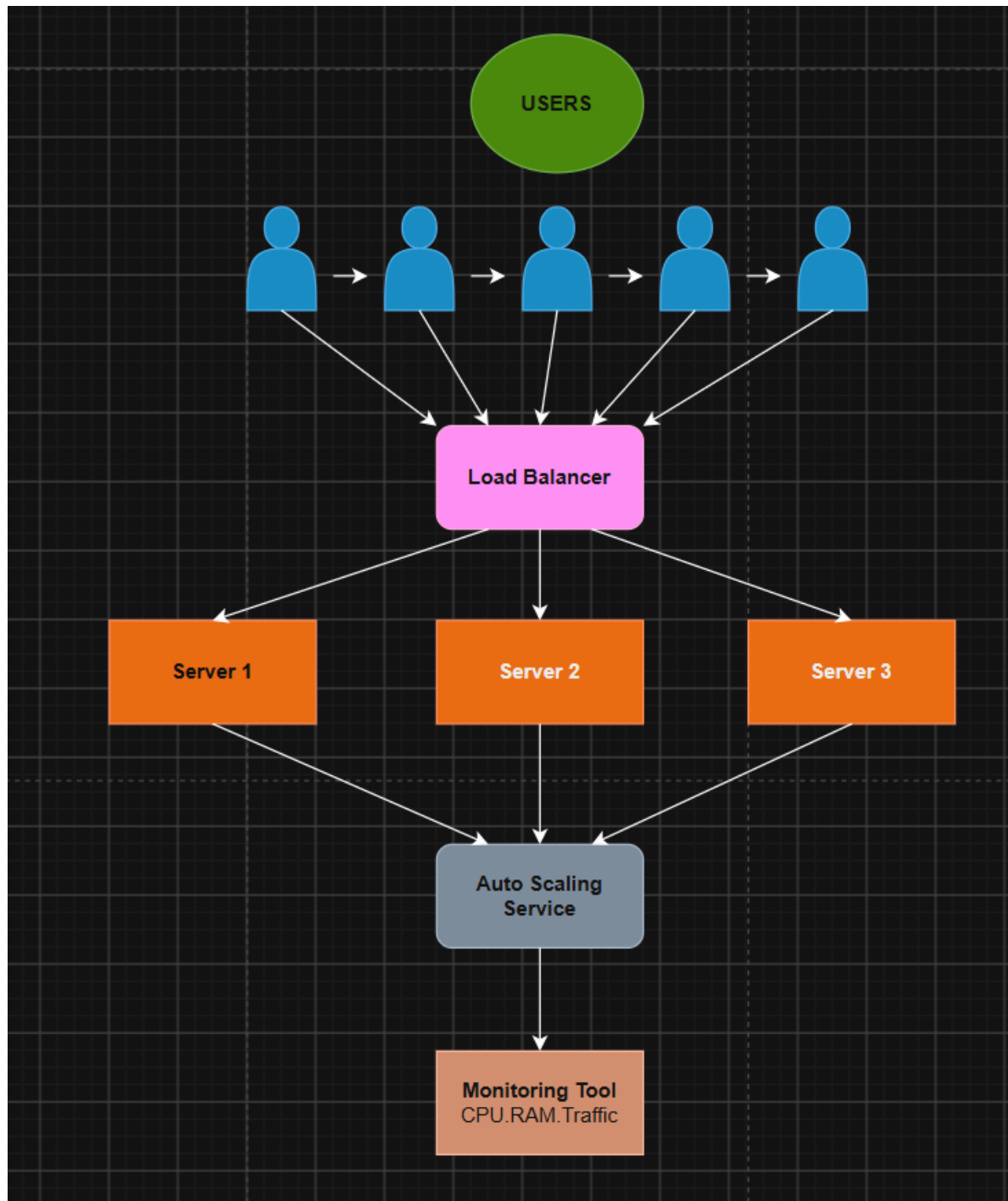
These two technologies together help reduce downtime. Downtime means the website is unavailable or not working properly. Since cloud resources can increase automatically and the load balancer redirects traffic to healthy servers, the chances of website failure become very low. Customers can continue shopping without facing errors or long waiting times.

This also improves the user experience. Customers can open pages quickly, search for products faster, add items to the cart easily and complete payments without delays. A fast and smooth website makes customers happy and increases the company's sales.

Monitoring tools play an important role in this setup. They continuously check CPU usage, memory usage, network traffic, server health and the number of users visiting the website. If the monitoring system detects high resource usage, it automatically informs the Auto Scaling service to launch additional servers. It also sends alerts to the system administrator if any problem occurs, so it can be solved immediately before it affects customers.

A good real-world example is Amazon or Flipkart during the Great Indian Festival or Big Billion Days sale. Millions of people visit these websites at the same time. Cloud Auto Scaling automatically adds more servers to handle the increased traffic. Load Balancing distributes customer requests across all available servers, while monitoring tools continuously watch system performance and generate alerts if needed. Because of these cloud services, customers can browse products, place orders and make payments smoothly even during peak shopping hours.

In conclusion, cloud Auto Scaling, Load Balancing and Monitoring Tools work together to handle heavy traffic efficiently. They reduce downtime, improve website performance, provide a better user experience and help companies save costs by using cloud resources only when required.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

ANS:

Nowadays organizations generate a large amount of important data every day. This data includes customer records, employee information, financial details and project files. Losing this data can create serious problems for the company. Therefore, many organizations use cloud storage to store backup data safely. Cloud storage is more reliable than traditional storage because it keeps multiple copies of the same data in different locations. If one server fails due to hardware problems or natural disasters, another copy of the data is available. This process is called data redundancy, and it helps prevent data loss.

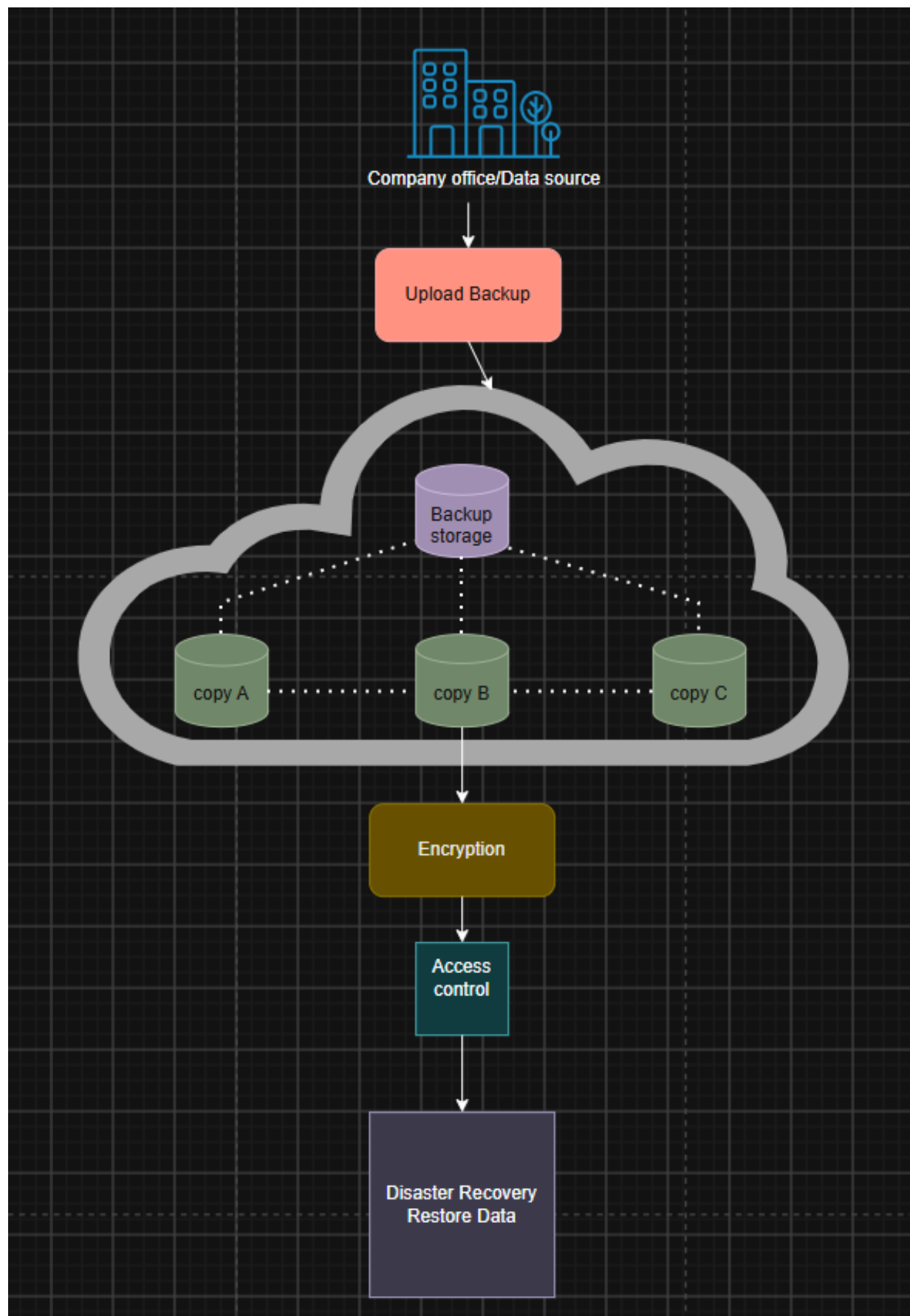
Cloud providers also provide high durability, which means the stored data remains safe for a long time. They regularly check the health of storage devices and automatically replace damaged hardware without affecting the stored information. This makes cloud storage more dependable than storing data only on local computers or office servers.

Security is another important advantage of cloud storage. The backup files are protected using encryption. Encryption converts the original data into an unreadable format so that only authorized users with the correct key can read it. Even if someone steals the data, they cannot understand it without the encryption key.

Cloud storage also uses access control to protect sensitive information. Only authorized users are allowed to view or modify the backup files. Companies can provide different permissions to employees depending on their job roles. Multi-factor authentication can also be used to improve security by asking for an additional verification code during login.

Cloud storage is also more economical than traditional storage systems. Companies do not need to purchase expensive hard disks, servers or backup devices. They also save money on maintenance, electricity and hardware replacement. Since cloud providers follow a pay-as-you-use model, organizations pay only for the storage they actually use. A good example of disaster recovery is a company whose office is damaged because of a flood or fire. Even if all local computers stop working, the backup stored in the cloud remains safe. The company can quickly restore its important files and continue its business without losing valuable information.

In conclusion, cloud storage provides secure, durable and cost-effective backup solutions. It protects important business data through redundancy, encryption and access control while supporting fast disaster recovery.



3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

ANS:

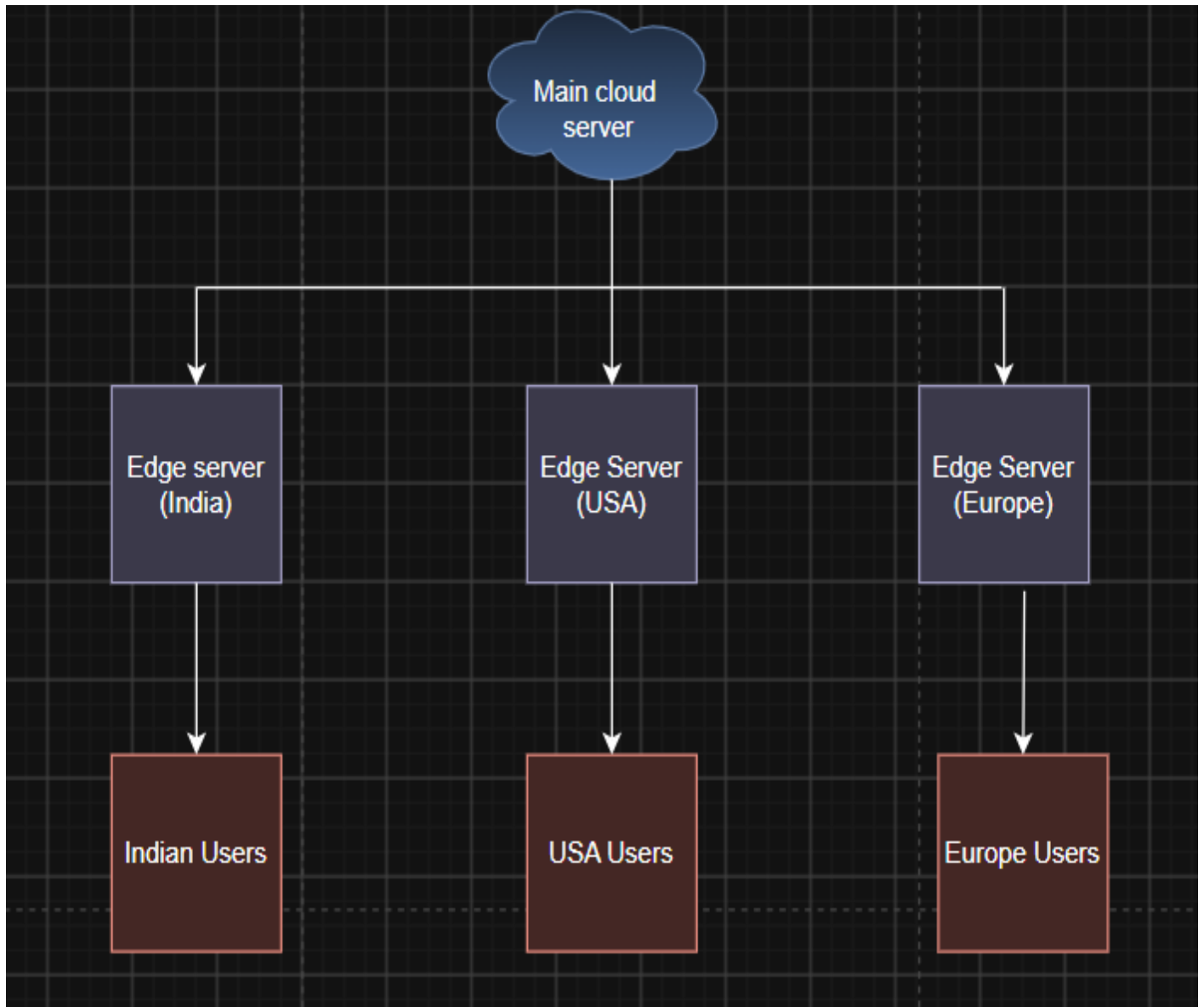
Many mobile applications are used by people from different countries. If the application is hosted on only one server, users who are far away may experience slow loading because the data has to travel a long distance. Cloud providers solve this problem by using a Content Delivery Network (CDN).

A CDN is a network of servers located in different parts of the world. Instead of sending data from one central server, the CDN stores copies of frequently used content like images, videos, web pages and application files on servers located near users. When a user opens the application, the nearest server sends the required data. This reduces the time taken to load the application and improves its speed.

The servers located near users are called edge servers. These servers reduce network delay, also called latency. Since the data travels a shorter distance, users experience faster loading, smooth video streaming and better application performance. Edge servers also reduce the workload on the main server because requests are shared among many servers. Cloud providers such as AWS, Microsoft Azure and Google Cloud have data centers in many countries. This allows applications to remain available even if one data center experiences a failure. Traffic is automatically redirected to another available location, ensuring continuous service.

A real-life example is Netflix. When people watch movies or web series, the video is delivered from the nearest CDN server instead of a distant server. This reduces buffering and provides better video quality. Similarly, online games like PUBG or Free Fire use nearby servers to reduce delay and improve the gaming experience.

In conclusion, CDNs and edge servers improve application speed, reduce latency and provide better performance for users across the world.



4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

ANS:

Financial institutions such as banks and insurance companies handle highly confidential customer information. Therefore, they require strong security and strict compliance while using cloud services. Cloud providers help organizations meet government regulations and international security standards by providing secure infrastructure and regular auditing facilities.

Cloud providers follow standards such as ISO 27001, PCI-DSS and GDPR to ensure that customer information is protected. They also perform regular audits to verify that security rules are being followed correctly. These audits help organizations maintain trust and satisfy legal requirements.

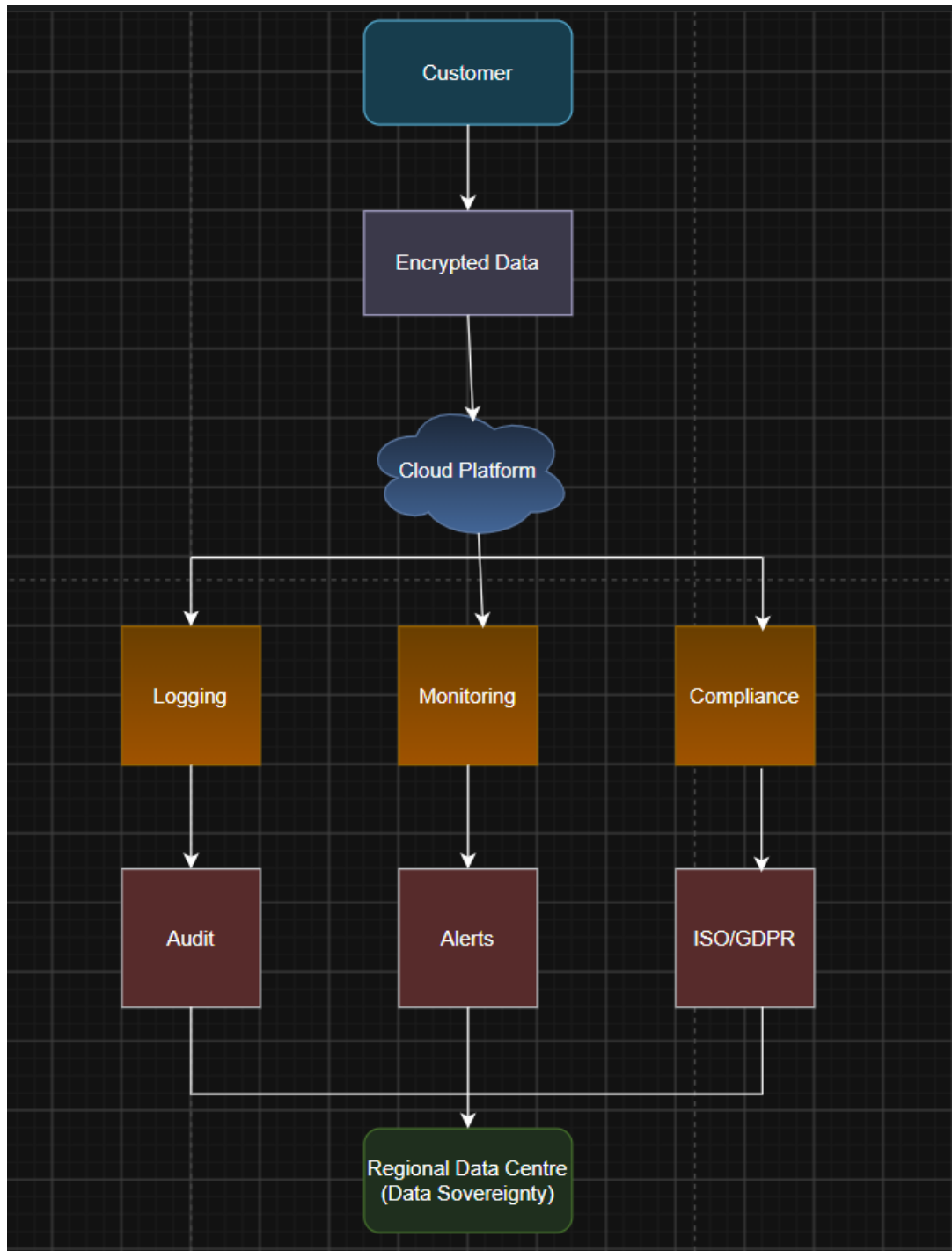
Another important security feature is logging. Every important activity, such as user login, money transfer, file access and account modification, is recorded in log files. These logs help administrators identify suspicious activities and investigate security incidents. Monitoring continuously checks the cloud environment for unusual behaviour. If someone tries to access customer data without permission or if abnormal activity is detected, the monitoring system immediately sends an alert so that action can be taken quickly.

Encryption is used to protect financial information both during transmission and while it is stored in the cloud. Even if attackers intercept the data, they cannot read it because it is encrypted.

Many countries also have data sovereignty laws, which require customer data to remain within the country's borders. Cloud providers allow organizations to choose specific data center regions so that data is stored only in approved locations and legal requirements are satisfied.

For example, when a customer transfers money through an online banking application, the transaction details are encrypted before being sent over the internet. The transaction is recorded in log files, monitored continuously for fraud and stored securely in the selected cloud region. This ensures both security and legal compliance.

In conclusion, cloud providers offer strong security through compliance, auditing, logging, monitoring and encryption, making cloud computing suitable for financial organizations.



5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANS:

Many software companies now use containerization because it makes application deployment faster and more efficient. Containers are lightweight packages that contain the application along with all the required software and libraries. Unlike virtual machines, containers share the operating system, so they require less memory and storage.

Virtual machines need a separate operating system for every instance, which increases resource usage and startup time. Containers start within a few seconds and use fewer system resources. Because of this, more containers can run on the same server compared to virtual machines.

Managing many containers manually is difficult, so companies use Kubernetes, which is a container orchestration platform. Kubernetes automatically deploys containers, distributes traffic, restarts failed containers and increases or decreases the number of containers depending on user demand. This reduces manual work and improves application reliability.

Another important advantage of containers is portability. The same container can run on AWS, Microsoft Azure, Google Cloud or even on a local computer without changing the application. This makes migration between cloud platforms easy and saves development time.

A good example of a microservices application is an online shopping website. Instead of keeping everything in one program, different services such as user login, product catalog, shopping cart, payment, order processing and notifications are placed in separate containers. If one service needs an update, only that container is modified without affecting the other services.

In conclusion, containerization improves application performance, reduces resource usage and makes deployment easier. Kubernetes simplifies container management, while portability allows applications to work across different cloud environments.

